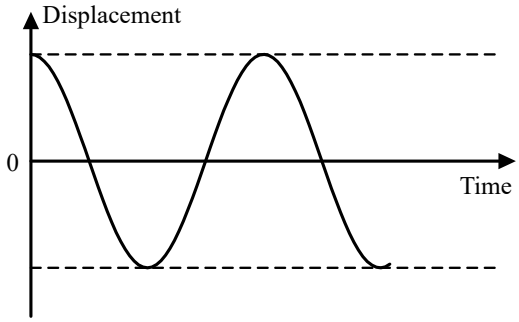
STPM 2014 PHYSICS 960/3 (U3)

Questions and Answers

Values of constants: $h = 6.63 \times 10^{-34}$ J s, $c = 3.00 \times 10^{8}$ m s^{-1}, $e = 1.60 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg, 1 u $= 1.66 \times 10^{-27}$ kg.

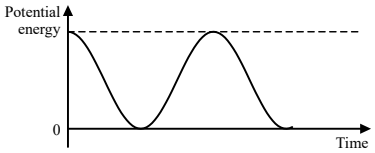
Section A [15 marks]

1. The displacement-time graph for an object moving in a simple harmonic motion is shown below.

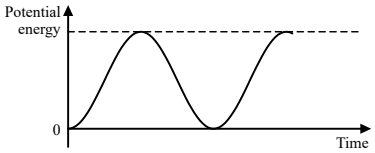


Which graph represents the variation of potential energy with time?

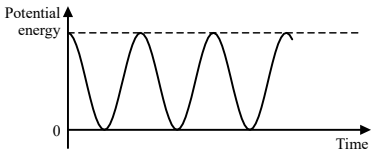
A



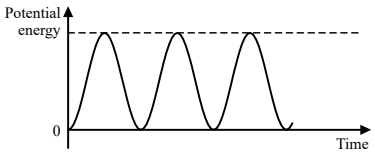
B



C



D



Answer: C

The displacement starts at its maximum, so the potential energy ($\propto x^2$) starts at its maximum and repeats twice per oscillation: graph C.

Explanation: Potential energy is never negative and has half the period of the displacement.

2. Which statement is **not** true when an oscillatory system is in critical damping?

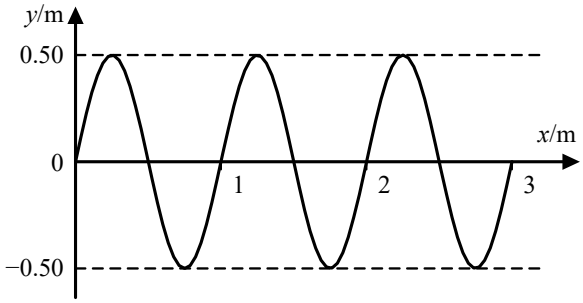
A The displacement decreases with time.
B The system will stop after a few oscillations.
C The system will stop in the shortest possible time.
D The total energy of the system decreases with time.

Answer: B

A critically damped system does not oscillate at all; it returns to equilibrium in the shortest time. Statement B is not true.

Explanation: Energy is lost and the displacement decreases throughout.

3. The variation of displacement of particles at time $t = 0$ s when a progressive wave is moving at speed 10 m s^{-1} in the positive-x direction is shown in the graph below.



Which is the equation of the wave?

A $y = 0.5 \sin (20\pi t - 2\pi x + \pi)$

B $y = 0.5 \sin (20\pi t + 2\pi x + \pi)$

C $y = 0.5 \sin (20\pi t - 2\pi x + \frac{\pi}{2})$

D $y = 0.5 \sin (20\pi t + 2\pi x + \frac{\pi}{2})$

Answer: A

$$\lambda = 1.0 \text{ m} \Longrightarrow k = 2\pi; f = \frac{v}{\lambda} = 10 \text{ Hz} \Longrightarrow \omega = 20\pi$$

At $t = 0$ the graph is $y = 0.5 \sin 2\pi x$; with $-2\pi x$ (travelling in $+x$) this needs a phase constant π: $\sin (\pi - 2\pi x) = \sin 2\pi x$ (A).

Explanation: Check the sign of the x term for the direction of travel, then the phase constant from the $t = 0$ graph.

4. The electrical oscillations of an electromagnetic wave is given by $E = E_0 \sin (6 \times 10^8\ \pi t - 2\pi x)$, where x in metres and t in seconds. Which statement is true of the electromagnetic wave?

A The intensity of the wave is E_0.

B The wavelength of the wave is 1.0 m.

C The frequency of the wave is 6×10^8 Hz.

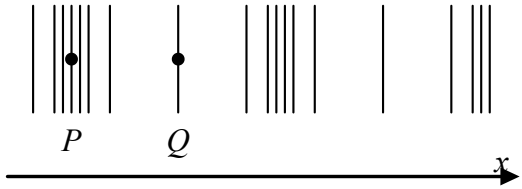
D The magnetic oscillations of the wave is $B = B_0 \sin (6 \times 10^8\ \pi t - 2\pi x + \frac{\pi}{2})$.

Answer: B

$$k = 2\pi \text{ m}^{-1} \Longrightarrow \lambda = 1.0 \text{ m}$$

Explanation: The frequency is 3×10^8 Hz (not 6×10^8), the intensity is proportional to E_0^2, and **B** is in phase with **E**.

5. The compressions and rarefactions of a sound wave travelling in air in positive-x direction is shown in the diagram below.



Which statement is true about the diagram above?

A At P, the air pressure is minimum.
B At Q, the air pressure is maximum.
C At Q, the air molecule is momentarily at rest.
D At P, the air molecule is moving in the positive-x direction.

Answer: D

P is at the centre of a compression. For a wave travelling in $+x$, air molecules at a compression are moving in the direction of the wave ($+x$).

Explanation: At a compression the pressure is maximum; Q is at a rarefaction (minimum pressure), where molecules move fastest in the $-x$ direction.

6. A car is moving towards an approaching ambulance. The car driver hears the siren of the ambulance. Which statement is true about the siren heard by the car driver?

A The apparent frequency is lower than the actual frequency.
B The apparent frequency is the same as the actual frequency.
C The apparent wavelength is the same as the actual wavelength.
D The apparent wavelength is shorter than the actual wavelength.

Answer: D

The ambulance (source) approaches, so the wavelength in front of it is shortened; the car moving towards it also raises the frequency heard.

Explanation: The apparent frequency is higher than the actual frequency.

7. A spherical mirror forms an upright image which is twice the size of the object. If the distance of object is 20 cm, what is the radius of curvature and the type of the mirror?

	Radius of curvature	*Type of mirror*
A	27 cm	Concave
B	27 cm	Convex
C	80 cm	Concave
D	80 cm	Convex

Answer: C

Upright and magnified $\Longrightarrow$ concave mirror, virtual image: $v = -40$ cm; $\frac{1}{f} = \frac{1}{20} - \frac{1}{40} \Longrightarrow f = 40$ cm, $r = 80$ cm

Explanation: Only a concave mirror can form a magnified upright image.

8. The side-view mirror of a car is convex with a radius of curvature 10 m. Another vehicle is 15 m from the mirror. What is the linear magnification of image of the vehicle as seen from the side-view mirror?

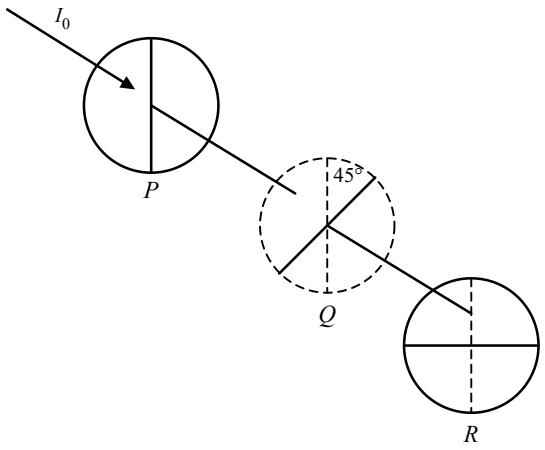
A 0.25 B 0.40 C 0.67 D 2.0

Answer: A

$$f = -5.0 \text{ m: } \frac{1}{v} = \frac{1}{-5.0} - \frac{1}{15} \Longrightarrow v = -3.75 \text{ m; } m = \frac{3.75}{15} = 0.25$$

Explanation: Convex mirrors give a wide field of view with a diminished image.

9. An unpolarised light of intensity I_0 is incident on two polarisers P and R with their polarising axes perpendicular to each other is shown in the diagram below.



A polariser Q with its polarising axis at 45° to that of P is then inserted in between P and R. What is the intensity of the light transmitted through R before and after the insertion of Q?

	Before	*After*
A	0	0
B	0	$\frac{1}{8}I_0$
C	$\frac{1}{4}I_0$	0
D	$\frac{1}{8}I_0$	0

Answer: B

Before: crossed polarisers transmit nothing: 0.

$$\text{After: } \frac{I_0}{2}\cos^2 45° \cos^2 45° = \frac{1}{8}I_0$$

Explanation: The middle polariser rotates the plane of polarisation so that some light gets through the last one.

10. A highly monochromatic beam could be transmitted in the optical fibre through

A refraction of the beam
B diffraction of the beam
C scattering of the laser beam
D total internal reflection of the beam

Answer: D

Light is guided along an optical fibre by total internal reflection.

Explanation: The core has a higher refractive index than the cladding.

11. If the probability for a radioisotope to decay per hour is 0.50, what is the half-life of the radioisotope?

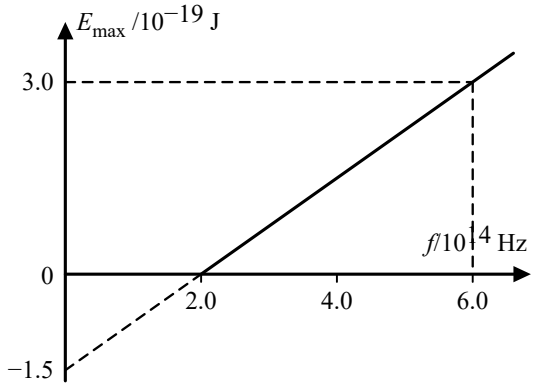
A 0.50 hour B 0.72 hour C 1.4 hour D 2.0 hour

Answer: C

$$\lambda = 0.50\ \text{h}^{-1} \Longrightarrow T_{½} = \frac{\ln 2}{0.50} = 1.4\ \text{h}$$

Explanation: The decay constant is the probability of decay per unit time.

12. The variation of maximum kinetic energy E_{max} of photoelectron with frequency f of light irradiated on the cathode in a photoelectric effect experiment is shown in the graph below.



Which deduction from the graph is **not** correct?

A The stopping potential is 0.94 V.
B The threshold frequency is 2.0×10^{14} Hz.
C The work function of the metal is 1.5×10^{-19} J.
D The Planck's constant obtained is 7.5×10^{-34} J s.

Answer: A

From the graph: threshold frequency 2.0×10^{14} Hz, work function 1.5×10^{-19} J (intercept), gradient $h = 3.0 \times 10^{-19}/4.0 \times 10^{14} = 7.5 \times 10^{-34}$ J s.

0.94 V is W/e, not a stopping potential; the stopping potential depends on the frequency. Statement A is not correct.

Explanation: $E_{max} = hf - W$: gradient h, intercept $-W$.

13. In a certain atom, the transition of electron from orbit R to orbit P produces a wavelength of 500 nm and the transition of electron from orbit Q to orbit P produces a wavelength of 600 nm. What wavelength will be produced for the transition from orbit R to orbit Q?

A 100 nm B 273 nm C 1100 nm D 3000 nm

Answer: D

$$\frac{hc}{\lambda_{RQ}} = \frac{hc}{500} - \frac{hc}{600} \Longrightarrow \lambda_{RQ} = 3000 \text{ nm}$$

Explanation: Energies of transitions add, not wavelengths.

14. The following features are the advantages of the use of nanoscience in electronic devices **except**

A smaller size devices
B faster processing of data
C higher consumption of energy
D lighter weight and thinner devices

Answer: C

Nanotechnology lowers, not raises, the energy consumption of devices. Option C is not an advantage.

Explanation: Smaller, faster, lighter devices are advantages.

15. What is the role of the control rod in producing a controlled nuclear chain reaction in a nuclear fission reactor?

A To absorb the excess neutrons
B To cool down the nuclear reactor
C To increase the rate of nuclear reaction
D To reduce the kinetic energy of neutrons

Answer: A

Control rods absorb excess neutrons to keep the chain reaction steady.

Explanation: The moderator reduces the kinetic energy of neutrons; the coolant removes heat.

Section B [15 marks]

16. (a) State **two** characteristics of a standing wave. [2]

Answer

• Nodes and antinodes at fixed positions — 1

• Amplitude varies with position; no net transfer of energy — 1

2

Explanation: Points between adjacent nodes oscillate in phase.

(b) The standing wave expression is represented by $y = 0.5 \cos (\${frac('\pi','4')})x \sin 80t$, where x and y are in metres and t in seconds. Determine the

(i) time interval for successive instances when the displacement of the wave is zero, [3]

Answer

$y = 0$ for all x when $\sin 80t = 0$ — 1

$80t = n\pi \Longrightarrow$ interval $= \frac{\pi}{80}$ — 1

$= 0.0393$ s (half the period, $T = 0.0785$ s) — 1

3

Explanation: The whole string is straight twice in every period.

(ii) distance between successive nodes. [3]

Answer

Nodes where $\cos (\pi x/4) = 0$ — 1

$k = \frac{\pi}{4} \Longrightarrow \lambda = 8.0$ m — 1

Distance between nodes $= \frac{\lambda}{2} = 4.0$ m — 1

3

Total for question **8**

Explanation: Successive nodes are half a wavelength apart.

17. (a) State **one** characteristic of an alpha particle. [1]

Answer

Any one: a helium nucleus (2 protons, 2 neutrons) / charge $+2e$ / strongly ionising / low penetration (stopped by paper) — 1

1

Explanation: It is the heaviest of the common radiations.

(b) An isotope of Polonium $^{218}_{84}$Po decays by emitting an alpha particle or a beta negative and energy to transform into the respective daughter nuclides of Lead Pb-214 and Astatine At-218.

(i) Write the complete reactions for the two possible decays. [2]

Answer

$^{218}_{84}\text{Po} \rightarrow {}^{214}_{82}\text{Pb} + {}^{4}_{2}\text{He} + Q$ 1

$^{218}_{84}\text{Po} \rightarrow {}^{218}_{85}\text{At} + {}^{0}_{-1}\text{e}$ (+ antineutrino) + Q 1

2

Explanation: Balance the nucleon and proton numbers.

(ii) Determine the energy released by each decay.
[Atomic mass of Po-218 = 218.008965 u, atomic mass of Pb-214 = 213.999798 u, atomic mass of At-218 = 218.008681 u and mass of alpha particle = 4.002603 u.] [4]

Answer

α: $\Delta m = 218.008965 - (213.999798 + 4.002603) = 0.006564$ u 1

$E = (0.006564)(1.66 \times 10^{-27})(3.00 \times 10^{8})^2 = 9.81 \times 10^{-13}$ J (6.1 MeV) 1

β: $\Delta m = 218.008965 - 218.008681 = 0.000284$ u (atomic masses include the electron) 1

$E = 4.24 \times 10^{-14}$ J (0.26 MeV) 1

4

Total for question **7**

Explanation: With atomic masses, the β^- electron is already accounted for by the extra orbital electron of At.

Section C [30 marks]

18. (a) Describe the propagation of sound in air. [2]

Answer

Sound is a longitudinal wave: air molecules vibrate about their mean positions parallel to the direction of propagation 1

producing compressions and rarefactions that travel through the air 1

2

Explanation: Energy is transferred, not the air itself.

(b) A point source of sound produces air vibration of amplitude 2.0×10^{-9} m at a distance of 5.0 m away. The threshold amplitude for hearing is 1.1×10^{-11} m.

(i) Calculate the sound intensity level at that position. [3]

Answer

$I \propto A^2 \Longrightarrow \beta = 10 \log \left(\frac{A}{A_0}\right)^2$ 1

$= 20 \log \frac{2.0 \times 10^{-9}}{1.1 \times 10^{-11}}$ 1

$= 45.2$ dB 1

3

Explanation: The threshold amplitude corresponds to 0 dB.

(ii) Determine the minimum distance from the source at which the sound could not be heard. [2]

Answer

$I \propto 1/r^2$ and $I \propto A^2 \Longrightarrow A \propto 1/r$: $r = 5.0 \times \frac{2.0 \times 10^{-9}}{1.1 \times 10^{-11}}$ 1

$r = 909$ m 1

2

Explanation: Beyond this distance the amplitude is below the threshold of hearing.

(c) A piano string of mass 9.00 g and length 62.5 cm is stretched to a tension of 900 N. The speed of wave v in a stretched string is given by $v = \sqrt{\frac{T}{\mu}}$, where T is the tension and μ is the mass per unit length of the string. The string produces standing waves when struck.

(i) Calculate the wavelength of the fundamental mode in the string. [2]

Answer

Fundamental: $L = \frac{\lambda}{2} \Longrightarrow \lambda = 2(0.625)$ 1

$\lambda = 1.25$ m 1

2

Explanation: Nodes at both fixed ends.

(ii) Calculate the fundamental frequency of the string vibration. [2]

Answer

$\mu = \frac{9.00 \times 10^{-3}}{0.625} = 0.0144 \text{ kg m}^{-1}$; $v = \sqrt{(\frac{900}{0.0144})} = 250 \text{ m s}^{-1}$ 1

$f = \frac{250}{1.25} = 200 \text{ Hz}$ 1

2

Explanation: The speed on the string is set by its tension and mass per unit length.

(iii) Explain why the sound produced by the string vibration has the same frequency but different wavelength from the wave in the string. [2]

Answer

The string forces the air to vibrate at the same frequency (the frequency is set by the source) 1

The speed of sound in air (about 340 m s^{-1}) differs from the wave speed on the string, so $\lambda = v/f$ is different 1

2

Explanation: Frequency does not change when a wave passes into a different medium.

(d) A tuning fork of frequency 440 Hz is sounded at the same time a piano note is played. Two beats per second is heard. The piano string is then tightened slightly, and the process is repeated. Three beats per second is heard. Determine the final frequency of the piano note. [2]

Answer

Initially 438 Hz or 442 Hz; tightening raises the frequency and the beats increase from 2 to 3, so it was 442 Hz 1

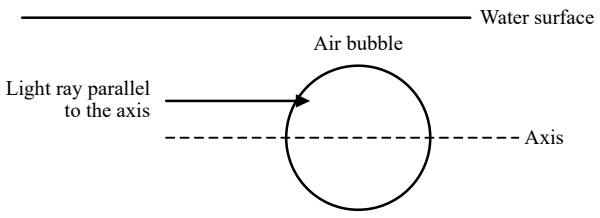
Final frequency = 443 Hz 1

2

Total for question **15**

Explanation: If it had been 438 Hz, tightening would have reduced the beat frequency.

19. (a) A spherical air bubble of radius 4.0 mm in water is shown in the diagram below.



A light ray parallel to the axis enters into the air bubble. The refractive index of water is 1.33.

(i) Determine the position of image due to refraction at the first surface. [3]

Answer

$\frac{n_1}{u} + \frac{n_2}{v} = \frac{n_2 - n_1}{r}$ with $u = \infty$, $n_1 = 1.33$, $n_2 = 1$, $r = +4.0$ mm — 1

$\frac{1}{v} = \frac{1 - 1.33}{4.0}$ — 1

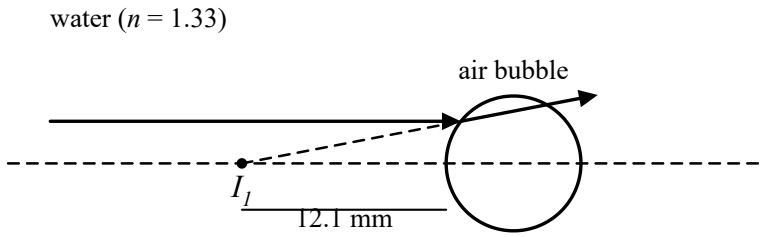
$v = -12.1$ mm: a virtual image 12.1 mm in front of the first surface — 1

3

Explanation: An air bubble in water acts as a diverging lens.

(ii) Sketch a ray diagram showing the position of the image formed in (a)(i). [1]

Answer



Refracted ray diverging, its extension meeting the axis at the virtual image 12.1 mm before the surface — 1

1

Explanation: The ray bends away from the normal as it enters the less dense air.

(iii) Determine the position of the final image after refraction at the second surface. [3]

Answer

Object for the second surface: $u = 12.1 + 8.0 = 20.1$ mm (real); $n_1 = 1$, $n_2 = 1.33$, $r = -4.0$ mm — 1

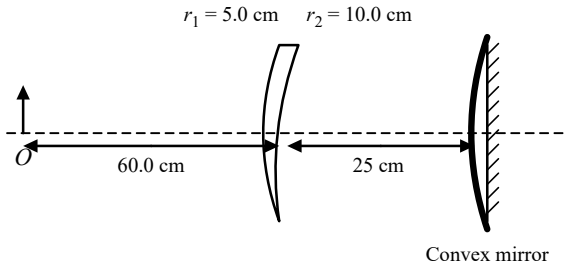
$\frac{1}{20.1} + \frac{1.33}{v} = \frac{1.33 - 1}{-4.0}$ — 1

$v = -10.1$ mm: final virtual image 10.1 mm from the second surface on the incident side — 1

3

Explanation: The second surface is concave towards the light inside the bubble, so its radius is negative.

(b) The arrangement of a lens and a convex mirror is shown in the diagram below.



The radii of curvatures of the lens r_1 and r_2 are 5.0 cm and 10.0 cm respectively. The refractive index of the lens is 1.5. The convex mirror of focal length 10.0 cm is placed 25 cm at the back of the lens. An object O is placed 60.0 cm in front of the lens. Determine the

(i) focal length of the lens, [3]

Answer

$\frac{1}{f} = (n-1)(\frac{1}{r_1} - \frac{1}{r_2})$ for a meniscus lens — 1

$= 0.5(\frac{1}{5.0} - \frac{1}{10.0})$ — 1

$f = 20.0$ cm — 1

3

Explanation: Both surfaces curve the same way; the more strongly curved surface dominates, so the lens converges.

(ii) position of the image formed by the lens, [2]

Answer

$\frac{1}{v} = \frac{1}{20.0} - \frac{1}{60.0}$ — 1

$v = 30.0$ cm behind the lens — 1

2

Explanation: A real image would form here if the mirror were absent.

(iii) final position of the image formed by the lens and the convex mirror. [3]

Answer

The lens image is 30.0 − 25 = 5.0 cm behind the mirror: a virtual object, $u = -5.0$ cm — 1

$\frac{1}{v} = \frac{1}{-10.0} - \frac{1}{-5.0}$ — 1

$v = +10.0$ cm: a real image 10.0 cm in front of the convex mirror — 1

3

Total for question **15**

Explanation: Converging light reaching a convex mirror can still form a real image.

20. (a) Write the expression for the energy of a photon. [1]

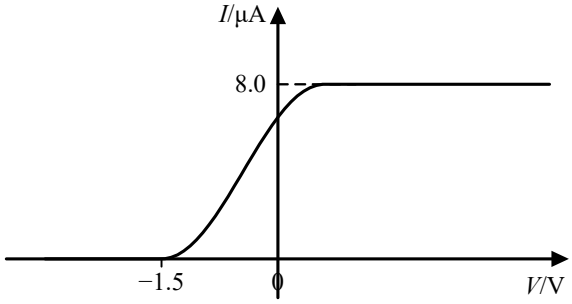
Answer

$E = hf = \frac{hc}{\lambda}$ 1

1

Explanation: h is the Planck constant.

(b) A graph of photocurrent I against the applied potential difference V in a photoelectric experiment is shown below.



The cathode is illuminated by a monochromatic light of frequency f.

(i) Determine the maximum kinetic energy of the photoelectrons emitted. [2]

Answer

From the graph $V_s = 1.5$ V: $K_{max} = eV_s$ 1

$= 1.5$ eV $= 2.4 \times 10^{-19}$ J 1

2

Explanation: The stopping potential is where the current falls to zero.

(ii) If the work function of the cathode is 1.02 eV, determine the frequency f of the light used. [3]

Answer

$hf = W + K_{max} = 1.02 + 1.5 = 2.52$ eV 1

$= 4.03 \times 10^{-19}$ J 1

$f = \frac{4.03 \times 10^{-19}}{6.63 \times 10^{-34}} = 6.08 \times 10^{14}$ Hz 1

3

Explanation: Einstein's photoelectric equation.

(iii) If only 20% of the incident photons emit photoelectrons, determine the power of the incident light. [4]

Answer

Electrons per second $= \frac{I_{sat}}{e} = \frac{8.0 \times 10^{-6}}{1.60 \times 10^{-19}} = 5.0 \times 10^{13}\ s^{-1}$ 1

Photons per second $= \frac{5.0 \times 10^{13}}{0.20} = 2.5 \times 10^{14}\ s^{-1}$ 1

$P = (2.5 \times 10^{14})(4.03 \times 10^{-19})$ 1

$P = 1.01 \times 10^{-4}$ W 1

4

Explanation: Use the saturation current, when all emitted electrons reach the anode.

(c)

(i) Name the phenomenon that shows wave nature of electrons. [1]

Answer

Electron diffraction 1

1

Explanation: Davisson and Germer showed it with a nickel crystal.

(ii) In a crystal, the successive layers of atoms are spaced at 1.15×10^{-10} m. Determine the angle of the crystal surface at which the first order Bragg diffraction for electrons of kinetic energy 200 eV occurs. [4]

Answer

$\lambda = \frac{h}{\sqrt{(2meV)}} = \frac{6.63 \times 10^{-34}}{\sqrt{(2(9.11 \times 10^{-31})(1.60 \times 10^{-19})(200))}}$ 1

$\lambda = 8.68 \times 10^{-11}$ m 1

$2d \sin\theta = \lambda \Longrightarrow \sin\theta = \frac{8.68 \times 10^{-11}}{2(1.15 \times 10^{-10})}$ 1

$\theta = 22.2°$ 1

4

Total for question **15**

Explanation: The electron wavelength is comparable with the atomic spacing, so diffraction is observed.